

prosail: an R package to simulate canopy reflectance with the coupled model PROSAIL (PROSPECT + SAIL)

Jean-Baptiste Féret¹✉ and Florian de Boissieu¹

¹ TETIS, INRAE, AgroParisTech, CIRAD, CNRS, Université Montpellier, Montpellier, France ✉
Corresponding author

DOI: [10.21105/joss.10451](https://doi.org/10.21105/joss.10451)

Software

- [Review](#) ✉
- [Repository](#) ✉
- [Archive](#) ✉

Editor: [Matthias Grenié](#) ✉

Reviewers:

- [@mmann1123](#)
- [@AlejandroP433](#)

Submitted: 31 January 2026

Published: 15 September 2026

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

Summary

PROSAIL is a vegetation radiative transfer model (RTM) coupling the *leaf optical PRO*PERTIES *SPECTra* (PROSPECT) model, which simulates leaf optical properties ([Jacquemoud & Baret, 1990](#)), with the *Scattering by Arbitrarily Inclined Leaves* model (SAIL) which simulates canopy bidirectional reflectance ([Berger et al., 2018](#); [Jacquemoud et al., 2009](#); [W. Verhoef, 1984](#)) from the visible to shortwave infrared domains. PROSAIL is a valuable tool for agriculture, ecology, and climate research ([Poulter et al., 2023](#)). Inversion procedures allow retrieval of vegetation biophysical properties, including leaf area index (LAI), leaf chemistry, the fraction of absorbed photosynthetically active radiation (fAPAR), and fractional vegetation cover (fCover).

We introduce *prosail*, an R package which provides multiple versions of the model PROSAIL, coupled *prospect* ([Féret & Boissieu, 2024](#)). *prosail* simulates surface reflectance in forward mode for various optical sensors, based on their spectral response function (SRF). It includes functions for model inversion using iterative optimization, and a hybrid approach, combining physical modeling and machine learning (ML) regression.

Statement of need

Vegetation traits mapping and monitoring is crucial to better understand ecosystem functions, carbon, water and energy budgets. Vegetation RTMs describe the interactions between light and vegetation based on their biophysical and chemical properties, through absorption and scattering mechanisms.

The SAIL model is an example of four-stream representations of the radiative transfer equation, including two direct fluxes (incident solar flux and radiance in the viewing direction) and two diffuse fluxes (upward and downward hemispherical flux).

More complex models include the *Soil Canopy Observation of Photosynthesis and Energy fluxes* (SCOPE) ([Yang et al., 2021](#)) model, which features layered description of the vegetation, and the *Discrete Anisotropic Radiative Transfer* DART model ([Gastellu-Etchegorry et al., 2015](#)), which explicitly describes 3D canopy structure.

Hybrid strategies generating a training dataset for ML algorithms with RTM simulations show multiple advantages ([Verrelst et al., 2015](#)): simulations complement or replace in situ sampling for robust estimation, while ML ensures computation efficiency compared to iterative optimization.

The limited number of input parameters and the computational efficiency of SAIL is interesting for operational applications, despite the main assumption of vegetation as a homogeneous

turbid medium: leaves are randomly distributed within the canopy, which is not accurate for row crops and heterogeneous canopies.

State of the field

Various softwares allow hybrid inversion with PROSAIL simulations. The *Sentinel Toolbox Application Platform* (SNAP) includes the Biophysical Processor module (Weiss et al., 2020), combining PROSAIL simulations with an artificial neural network regression model. The sampling design of the training set is described in the *Algorithmic Theoretical Background Document* (ATBD) (Weiss et al., 2020)). Alternative distributions provide more interactive parameterization of the inversion strategy, e.g. the *Automated Radiative Transfer Models Operator* (ARTMO) Matlab toolbox (Caicedo et al., 2014).

`prosail` does not intend to provide the same computational efficiency as SNAP. It does not provide a collection of models and methods as comprehensive as those provided with the ARTMO box neither. `prosail` provides a flexible open source framework to experiment with hybrid inversion procedures, using simple yet fast and efficient training stage, allowing experimenting on training data sampling design, introduction of noise, feature selection over optical sensors. It is suitable for research and development, to identify potential and limitations of inversion strategies. Resulting regression models can be applied on remote sensing data, but it may not be the appropriate software for scaling up operational vegetation monitoring applications. Alternative PROSAIL implementations are also available in [python](#), [Julia](#) and [R](#). Other options are available on [this webpage](#).

Software design

`prosail` uses `prospect` (Féret et al., 2017, 2021) for the simulation of leaf optical properties. It includes `4SAIL` (W. Verhoef et al., 2007), and `4SAIL2` (Wout Verhoef & Bach, 2007), a two layers version of `4SAIL` as canopy models. `prosail` provides user-friendly and modular functions to simulate vegetation canopy reflectance, and to predict biophysical properties using inversion.

Hybrid inversion is a multi-step procedure requiring simulation of sensor reflectance to train a ML regression algorithm, then applicable to any data source collected from airborne and spaceborne sensors.

- The generation of realistic reflectance simulations requires definition of input parameter sampling strategy, accounting for sensor SRF and uncertainties. `prosail` provides functions to produce this training dataset as a unique step following standard parameterization, or step by step, in order for users to understand the logic, or adjust each step to their needs.
- A limited number of ML algorithms is currently provided. Additional algorithms will be implemented in future versions.
- Raster processing is handled with the `terra` (Hijmans, 2025) package, for compatibility with most raster data formats.
- `prosail` can be used in combination with `preprocS2`, a package dedicated to downloading and preprocessing of Earth observation data following the [STAC](#) specification, in order to produce a fully automated image access and processing workflow.

Research impact statement

`prosail` has been used in multiple research publications since its early developments for applications in agriculture ((Verrelst et al., 2021)), ecology ((Hauser et al., 2021), (Féret et

al., 2026)), forestry (([Ferreira et al., 2026](#))), and vegetation monitoring in general ([Kattenborn et al., 2024](#)). The model PROSAIL is used to analyze optical imagery acquired from UAV to airborne to spaceborne sensors, including multispectral satellite (Sentinel-2, Landsat) and imaging spectroscopy sensors (EnMAP, PRISMA, AVIRIS). This package is currently used in multiple research projects, and is actively maintained.

Overview

Simulating canopy reflectance with PROSAIL

PROSAIL requires defining vegetation properties, including leaf optical properties simulated with prospect, or measured experimentally, LAI, and leaf inclination distribution. PROSAIL also requires information extrinsic to vegetation, including the foliage hot spot parameter ([Bréon et al., 2002](#); [Kuusk, 1991](#); [Wout Verhoef & Bach, 2007](#)), the sun-observer geometry (sun and observer zenith angles, relative azimuth angle), and soil reflectance. SAIL produces top-of-canopy bi-hemispherical reflectance factor, directional-hemispherical reflectance factor for solar incident flux, hemispherical-directional reflectance factor in viewing direction and bi-directional reflectance factor.

These reflectance factors can then be used to simulate surface reflectance acquired by a sensor with known SRF. The SRF from multiple sensors is already implemented in `prosail`, listed when calling `srf_availability()`. User-defined sensors require either defining central wavelength and full width at half maximum (`fwhm`) for each band, assuming gaussian response, or the exact SRF provided with a `.csv` files.

PROSAIL hybrid inversion

Simulating surface reflectance to prepare for regression model training

Regression models are trained based on i. a sampling strategy (range, distribution, co-distributions for input parameters), ii. a level of noise applicable to the simulated surface reflectance, and iii. spectral bands used for the training of the regression model.

The default parameterization of `prosail` hybrid inversion relies on the production of a surface reflectance LUT compliant with the specifications defined in the ATBD for the Biophysical Processor of the Sentinel toolbox ([Weiss et al., 2020](#)). Each step of the simulation of a training LUT can be defined by user.

Training a ML regression algorithm

The ML strategy implemented in `prosail` is a parsimonious ensemble method based on a bootstrap aggregating (bagging) prediction of biophysical properties from support vector regression (SVR) models. Each SVR model is trained with a limited number of samples, ensuring fast training stage. The predicted value corresponds to the mean prediction from a set of SVR models. The standard deviation is derived from this ensemble of predictors. Standard deviation may not provide accurate model uncertainty quantification ([Palmer et al., 2022](#)).

The default parameterization uses an ensemble of 10 SVR models trained with 200 samples each. The training then requires 2000 samples, to be compared to the 41472 simulations used for training models implemented in SNAP. This parsimonious ensemble method ensures fast training stage and retrieval performances similar to SNAP.

Performing full hybrid inversion

Figure 1 summarizes the workflow applied to perform hybrid inversion. The function `train_prosail_inversion` combines aforementioned steps to simulate surface reflectance

and train SVR models. Obtained regression models are then used to estimate vegetation biophysical properties from data tables or raster data.

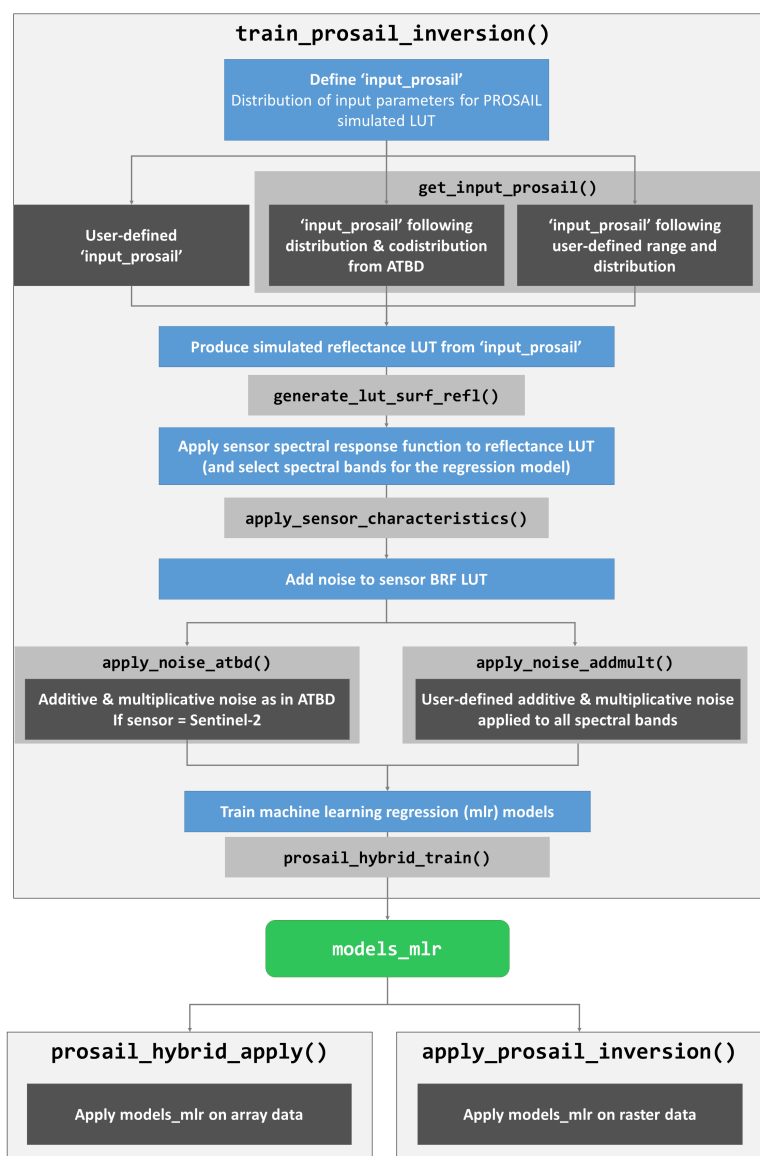


Figure 1: Workflow of PROSAIL hybrid inversion implemented in the package `prosail`.

Application to Sentinel-2 image and comparison with SNAP

Once ML regression models are trained, `prosail` hybrid inversion can be applied on Sentinel-2 imagery using the function `apply_prosail_inversion`. Biophysical variables produced with SNAP are compared to those produced with the `prosail` hybrid inversion. The Sentinel-2 Level-2A product corresponding to tile *30SWJ* acquired on *May 13th, 2021* over an agricultural landscape located next to Barrax (Spain) was downloaded from the Copernicus Data Space Ecosystem (CDSE). LAI, *fcover* and *fAPAR* computed from Sentinel-2 images with `prosail` hybrid inversion are compared to those produced with SNAP in Figure 2.

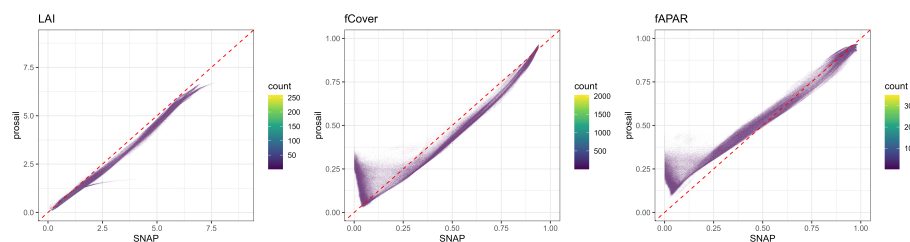


Figure 2: Biophysical properties estimated with *prosail* vs Biophysical properties estimated with SNAP.

The two methods show good consistency, with Pearson correlation coefficient > 0.99 for LAI, and > 0.97 for fCover and fAPAR. Algorithmic differences include the PROSPECT version (SNAP uses a version anterior to PROSPECT-D) and the ML algorithm. Differences in soil properties accounted for in simulations with low LAI may also contribute, despite efforts to reproduce the workflow described in the ATBD. Additional tests performed over croplands and forests showed similar performances. Figure 3 displays a set of biophysical properties mapped over the region of interest.

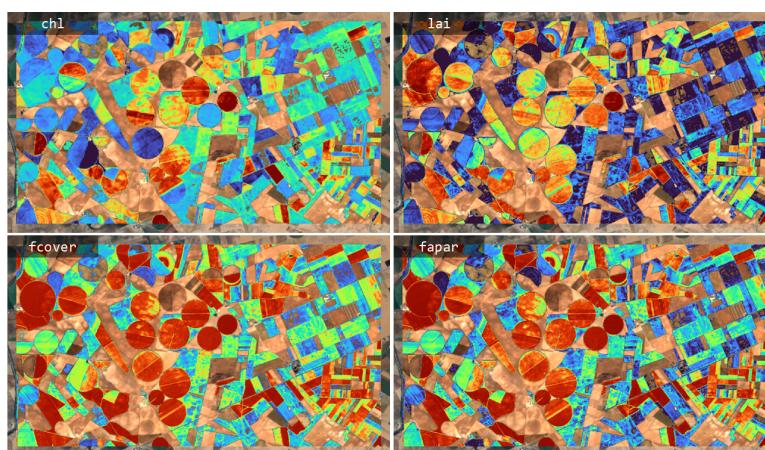


Figure 3: Leaf chlorophyll content, LAI, fCover and fAPAR mapped over the region of Barrax (Spain).

Conclusion

prosail is an R package dedicated to the canopy reflectance model PROSAIL, simulating reflectance from optical sensors based on their spectral response. It includes inversion based on iterative optimization and ML/RTM inversion. Vegetation biophysical properties estimated with a hybrid inversion are consistent with estimations from SNAP. *prosail* hybrid inversion does not intend to be computationally as efficient as SNAP. The current version not appropriate for regional to global scale vegetation monitoring. *prosail* hybrid inversion offers a fully adjustable hybrid inversion framework, including an original parsimonious ML regression method, allowing fast and efficient training. It is a valuable tool for experimenting on the potential and limitation of physically-based retrieval of vegetation traits from optical sensors, including satellite missions already operational and in preparation.

Availability

prosail is an open-source software made available under the GNU GPL-3 license. Tutorials are available at <https://jbferet.gitlab.io/prosail/>.

Acknowledgements

This work was supported in part by CNES through the TOSCA project SYLVOSANSAT, by the Agence Nationale de la Recherche (ANR) through the BioCop project (ANR-17-CE32-0001-01), and by government funding managed by ANR under the France 2030 program as part of the PEPR FORESTT, MONITOR, reference number ANR-24-PEFO-0003. We are grateful to Wout Verhoef for the 4SAIL and 4SAIL2 models. We are grateful to Stéphane Jacquemoud and Frédéric Baret for the initial version of the PROSPECT model. We are grateful to the reviewers who contributed to significantly improve the quality of the manuscript and the stability of this package.

AI Usage Disclosure

No generative AI tools were used in the development of this software, the writing of this manuscript, or the preparation of supporting materials.

References

- Berger, K., Atzberger, C., Danner, M., D'Urso, G., Mauser, W., Vuolo, F., & Hank, T. (2018). Evaluation of the PROSAIL model capabilities for future hyperspectral model environments: A review study. *Remote Sensing*, 10(1). <https://doi.org/10.3390/rs10010085>
- Bréon, F.-M., Maignan, F., Leroy, M., & Grant, I. (2002). Analysis of hot spot directional signatures measured from space. *Journal of Geophysical Research: Atmospheres*, 107(D16), AAC 1-1-AAC 1-15. <https://doi.org/10.1029/2001JD001094>
- Caicedo, J. P. R., Verrelst, J., Muñoz-Marí, J., Moreno, J., & Camps-Valls, G. (2014). Toward a semiautomatic machine learning retrieval of biophysical parameters. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 7(4), 1249–1259. <https://doi.org/10.1109/JSTARS.2014.2298752>
- Féret, J.-B., Berger, K., Boissieu, F. de, & Malenovsky, Z. (2021). PROSPECT-PRO for estimating content of nitrogen-containing leaf proteins and other carbon-based constituents. *Remote Sensing of Environment*, 252, 112173. <https://doi.org/10.1016/j.rse.2020.112173>
- Féret, J.-B., & Boissieu, F. de. (2024). Prospect: An r package to link leaf optical properties with their chemical and structural properties with the leaf model PROSPECT. *Journal of Open Source Software*, 9(94), 6027. <https://doi.org/10.21105/joss.06027>
- Féret, J.-B., Boissieu, F. de, & Lang, M. (2026). A plant trait-based approach to map forest biodiversity using imaging spectroscopy and physical modeling. In D. Rocchini (Ed.), *R coding for ecology* (pp. 161–182). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-99665-8_7
- Féret, J.-B., Gitelson, A. A., Noble, S. D., & Jacquemoud, S. (2017). PROSPECT-D: Towards modeling leaf optical properties through a complete lifecycle. *Remote Sensing of Environment*, 193, 204–215. <https://doi.org/10.1016/j.rse.2017.03.004>
- Ferreira, V. B., Féret, J.-B., Atzberger, C., Guillemot, J., Campoe, O., & Maire, G. le. (2026). Enhancing PROSAIL inversion: Key considerations and practical insights. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 19, 1378–1390. <https://doi.org/10.1109/JSTARS.2025.3635001>
- Gastellu-Etchegorry, J.-P., Yin, T., Lauret, N., Cajgfinger, T., Gregoire, T., Grau, E., Féret, J.-B., Lopes, M., Guilleux, J., Dedieu, G., Malenovsky, Z., Cook, B. D., Morton, D., Rubio, J., Durrieu, S., Cazanave, G., Martin, E., & Ristorcelli, T. (2015). Discrete anisotropic radiative transfer (DART 5) for modeling airborne and satellite spectroradiometer and

- LIDAR acquisitions of natural and urban landscapes. *Remote Sensing*, 7(2), 1667–1701. <https://doi.org/10.3390/rs70201667>
- Hauser, L. T., Féret, J.-B., An Binh, N., van der Windt, N., Sil, Â. F., Timmermans, J., Soudzilovskaia, N. A., & van Bodegom, P. M. (2021). Towards scalable estimation of plant functional diversity from sentinel-2: In-situ validation in a heterogeneous (semi-)natural landscape. *Remote Sensing of Environment*, 262, 112505. <https://doi.org/10.1016/j.rse.2021.112505>
- Hijmans, R. J. (2025). *Terra: Spatial data analysis*. <https://CRAN.R-project.org/package=terra>
- Jacquemoud, S., & Baret, F. (1990). PROSPECT: A model of leaf optical properties spectra. *Remote Sensing of Environment*, 34(2), 75–91. [https://doi.org/10.1016/0034-4257\(90\)90100-Z](https://doi.org/10.1016/0034-4257(90)90100-Z)
- Jacquemoud, S., Verhoef, W., Baret, F., Bacour, C., Zarco-Tejada, P. J., Asner, G. P., François, C., & Ustin, S. L. (2009). PROSPECT+SAIL models: A review of use for vegetation characterization. *Remote Sensing of Environment*, 113, S56–S66. <https://doi.org/10.1016/j.rse.2008.01.026>
- Kattenborn, T., Wieneke, S., Montero, D., Mahecha, M. D., Richter, R., Guimarães-Steinicke, C., Wirth, C., Ferlian, O., Feilhauer, H., Sachsenmaier, L., Eisenhauer, N., & Dechant, B. (2024). Temporal dynamics in vertical leaf angles can confound vegetation indices widely used in Earth observations. *Communications Earth & Environment*, 5(1), 550. <https://doi.org/10.1038/s43247-024-01712-0>
- Kuusk, A. (1991). The hot spot effect in plant canopy reflectance. In R. B. Myneni & J. Ross (Eds.), *Photon-vegetation interactions: Applications in optical remote sensing and plant ecology* (pp. 139–159). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-75389-3_5
- Palmer, G., Du, S., Politowicz, A., Emory, J. P., Yang, X., Gautam, A., Gupta, G., Li, Z., Jacobs, R., & Morgan, D. (2022). Calibration after bootstrap for accurate uncertainty quantification in regression models. *Npj Computational Materials*, 8(1), 115. <https://doi.org/10.1038/s41524-022-00794-8>
- Poulter, B., Currey, B., Calle, L., Shiklomanov, A. N., Amaral, C. H., Brookshire, E. N. J., Campbell, P., Chlus, A., Cawse-Nicholson, K., Huemmrich, F., Miller, C. E., Miner, K., Pierrat, Z., Raiho, A. M., Schimel, D., Serbin, S., Smith, W. K., Stavros, N., Stutz, J., ... Zhang, Z. (2023). Simulating global dynamic surface reflectances for imaging spectroscopy spaceborne missions: LPJ-PROSAIL. *Journal of Geophysical Research: Biogeosciences*, 128(1), e2022JG006935. <https://doi.org/10.1029/2022JG006935>
- Verhoef, W. (1984). Light scattering by leaf layers with application to canopy reflectance modeling: The SAIL model. *Remote Sensing of Environment*, 16(2), 125–141. [https://doi.org/10.1016/0034-4257\(84\)90057-9](https://doi.org/10.1016/0034-4257(84)90057-9)
- Verhoef, Wout, & Bach, H. (2007). Coupled soil–leaf–canopy and atmosphere radiative transfer modeling to simulate hyperspectral multi-angular surface reflectance and TOA radiance data. *Remote Sensing of Environment*, 109(2), 166–182. <https://doi.org/10.1016/j.rse.2006.12.013>
- Verhoef, W., Jia, L., Xiao, Q., & Su, Z. (2007). Unified optical-thermal four-stream radiative transfer theory for homogeneous vegetation canopies. *IEEE Transactions on Geoscience and Remote Sensing*, 45(6), 1808–1822. <https://doi.org/10.1109/TGRS.2007.895844>
- Verrelst, J., Camps-Valls, G., Muñoz-Marí, J., Rivera, J. P., Veroustraete, F., Clevers, J. G. P. W., & Moreno, J. (2015). Optical remote sensing and the retrieval of terrestrial vegetation bio-geophysical properties – a review. *ISPRS Journal of Photogrammetry and Remote Sensing*, 108, 273–290. <https://doi.org/10.1016/j.isprsjprs.2015.05.005>

- Verrelst, J., Rivera-Caicedo, J. P., Reyes-Muñoz, P., Morata, M., Amin, E., Tagliabue, G., Panigada, C., Hank, T., & Berger, K. (2021). Mapping landscape canopy nitrogen content from space using PRISMA data. *ISPRS Journal of Photogrammetry and Remote Sensing*, 178, 382–395. <https://doi.org/10.1016/j.isprsjprs.2021.06.017>
- Weiss, M., Baret, F., & Jay, S. (2020). *S2ToolBox Level 2 products LAI, FAPAR, FCOVER* [Research Report]. EMMAH-CAPTE, INRAe Avignon. <https://hal.inrae.fr/hal-03584016>
- Yang, P., Prikaziuk, E., Verhoef, W., & Tol, C. van der. (2021). SCOPE 2.0: A model to simulate vegetated land surface fluxes and satellite signals. *Geoscientific Model Development*, 14(7), 4697–4712. <https://doi.org/10.5194/gmd-14-4697-2021>